

Solution :

3. In case of $\int PdV$ and $\int VdP$, integral can be evaluated, when P can be expressed in terms of V (or V in terms of P). It is not an exact differential and hence not a property. In case of
- $$\int PdV + \int VdP = \int (PdV + VdP)$$
- $$= \int d(PV) = PV$$

So it can be evaluated without knowing a relation between P & V. It's an exact differential and hence is a thermodynamic a property.

4. For any reversible process.

$$\Delta S_{\text{system}} + \Delta S_{\text{surrounding}} = 0$$

5. No spontaneous heat transfer from a cold body to a hot body is possible.

6. $ds = \frac{dQ}{T} \Rightarrow \frac{ds}{dT} = \frac{nC}{T}$

$$\Rightarrow \frac{dT}{ds} = \frac{T}{nC} = \text{slope of T-S graph.}$$

$$\therefore \frac{\text{Slope of isobaric curve}}{\text{Slope of isochoric curve}} = \frac{\frac{T}{nC_p}}{\frac{T}{nC_v}} = \frac{1}{\gamma}$$

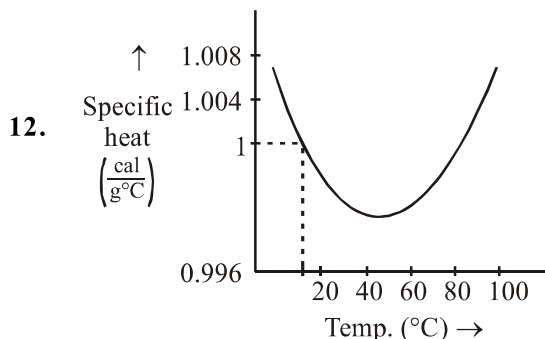
7. $\eta_{AB} = 1 - \frac{T_B}{T_A}; \eta_{AC} = 1 - \frac{T_C}{T_A}; \eta_{AD} = 1 - \frac{T_D}{T_A}$

As per conditions given.

$$\eta_{AC} = \frac{1}{2}(\eta_{AB} + \eta_{AD})$$

$$\Rightarrow 1 - \frac{T_C}{T_A} = 1 - \left(\frac{T_B + T_D}{2T_A} \right) \Rightarrow T_C = \frac{T_B + T_D}{2}$$

8. Internal energy does not change in isothermal process. ΔS can be zero for adiabatic process. Work done in an adiabatic process may be non-zero.
9. Efficiency depends on the temperature of source and sink.
10. Molar specific heat capacity of a substance is the amount of heat required to raise the temperature of one mole of that substance by unity.
11. Heat and work in thermodynamics are not state variables. The internal energy of a given mass of gas depends on its state described by specific values of pressure, volume and temperature.



13. $\eta = 1 - \frac{T_2}{T_1}$

$Q_1 = W + Q_2$ (heat release to the cold reservoir can never be made zero)

and for a refrigerator $\beta = \frac{Q_2}{W}$ (external work can never be zero)

14. • In this Process $P \propto V$ or $P = KV$
 • $V \uparrow \Rightarrow dV = \oplus \text{ve} \Rightarrow \Delta W = PdV = \oplus \text{ve}$
 • $PV \propto T$ and (PV) product increases then T also increases
 • $T \uparrow \Rightarrow dU = \oplus \text{ve}$, So, $\Delta Q = \Delta W + dU$
15. * The variables that remain unchanged for each part are intensive variables
 * The variables whose values yet halved in each part are extensive variables
16. Between 0°C to 4°C in water, volume decreases $V \downarrow \Rightarrow \Delta W = \ominus \text{ve}$ and $\Delta Q < dU$, So, $C_p < C_v$
17. * In adiabatic compression
 $Q = 0$ & $TV^{\gamma-1} = \text{constant}$ than $V \downarrow \Rightarrow T \uparrow$
 * $V \uparrow \Rightarrow dV = \oplus \text{ve} \Rightarrow W = PdV = \oplus \text{ve}$
 * Free expansion of ideal gas is isothermal expansion process but no heat is exchanged between system and surrounding
 * Work is a path function, not a state function
18. AB part $\Rightarrow V = \text{const.} \Rightarrow W = 0 \Rightarrow Q = dU$
 $P \propto T \Rightarrow P \uparrow \Rightarrow T \uparrow \Rightarrow \Delta U = +\text{ve}$
 & $\Delta Q = +\text{ve}$
 BC part $\Rightarrow Q = 0 \Rightarrow W = -dU$
 $TV^{\gamma-1} = \text{const.} \Rightarrow V \uparrow \Rightarrow T \downarrow \Rightarrow U \downarrow$
 CD part $\Rightarrow V = \text{const.} \Rightarrow W = 0 \Rightarrow Q = \Delta U$
 $P \propto T \Rightarrow P \downarrow \Rightarrow T \downarrow \Rightarrow \Delta U = -\text{ve}$
 & $\Delta Q = \ominus \text{ve}$
 DA part $\Rightarrow Q = 0 \Rightarrow W = -\Delta U$
 $TV^{\gamma-1} = \text{const.} \Rightarrow V \downarrow \Rightarrow T \uparrow \Rightarrow U \uparrow$
25. The water will be slightly warmer after the stirring than before, hence temperature $\uparrow$, $U \uparrow$.